



bigbasket

**Big basket with
(EDA)**

**Exploratory data analysis
Capstone project report**

Introduction to Big Basket

Big Basket is one of India's largest online grocery stores, offering a wide range of products from fresh produce to packaged food and other essentials.

This analysis aims to explore product data from Big Basket, focusing on various factors such as price, categories, and consumer ratings.

Key Offerings of Big Basket

1.Variety of Products: Big Basket offers a wide range of products, from fresh fruits and vegetables to packaged foods, beverages, personal care, household items, and even gourmet and organic food products.

2.Fast Delivery: One of the company's unique selling points is its same-day or next-day delivery service, providing convenience to customers who may need their groceries urgently.

3.Discounts and Offers: Big Basket often runs promotions, discounts, and offers to attract price-conscious shoppers. These deals, combined with convenient shopping options, have helped build its customer base.

The E-Commerce Advantage

Increased Digital Adoption: The rise of smartphones and internet access has made online grocery shopping more popular. Big Basket is one of the prominent platforms benefiting from this shift in consumer behavior.

Consumer Preferences: As people look for contactless shopping options and convenience, online grocery stores like Big Basket have seen rapid growth. Customers are more willing to experiment with online grocery shopping, especially due to the increasing quality of services and product ranges offered.

Target Audience

Big Basket's core customers are urban dwellers, often in the age group of 25-45, who value convenience, variety, and quality in their grocery purchases. They are typically working professionals, homemakers, or anyone looking for a seamless grocery shopping experience.

The platform has a pan-India presence, making it accessible to millions of customers across metro cities, tier-1, and tier-2 locations.

Importance of EDA in Big Basket's Data

To understand customer behavior and optimize the platform's offerings, Big Basket collects extensive data about products, categories, pricing, sales, and consumer preferences.

Exploratory Data Analysis (EDA) helps uncover trends, identify patterns, and improve decision-making in areas like:

- . Pricing strategies (Sale vs. Market Price)
- . Popularity of categories and subcategories
- . Brand performance
- . Customer sentiment analysis through product ratings

Purpose of This Analysis:

This presentation uses Exploratory Data Analysis (EDA) to analyze product data from Big Basket, focusing on key attributes such as pricing, product categories, brands, and ratings. The goal is to extract actionable insights that can support better decision-making and improve Big Basket's product offerings and

Overview of data

. The dataset contains information about products listed on Big Basket.

- . **index** - Simply the Index!
- . **Product** - Title of the product (as they're listed)
- . **category** - Category into which product has been classified
- . **subcategory** - Subcategory into which product has been kept
- . **brand** - Brand of the product
- . **sale price** - Price at which product is being sold on the site
- . **type** - Type into which product falls
- . **market price** - Market price of the product
- . **rating** - Rating the product has got from its consumers
- . **description** - Description of the dataset (in detail)

DATASET REPORTS

-Load data set and data exploration

First of all, we import our library

```
import pandas as pd
import NumPy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
```

Then after that we mount our drive

```
from google.colab import drive
drive.mount('/content/drive')
Mounted at /content/drive
```

-Libraries used

Pandas: Data Handling and Manipulation

Pandas is an open-source library that provides high-performance data structures like Data Frame and Series for data manipulation and analysis.

Key Features:

Data Frame: A two-dimensional labeled data structure, similar to a table or spreadsheet, which allows you to store and manipulate data.

Data Import/Export: Easily read and write data to/from CSV, Excel, SQL databases, and other file formats.

Data Cleaning: Handle missing values, duplicate

data, and apply transformations like filtering, grouping



and merging dataset.

Aggregation: Allows you to group data and perform operations like sum, mean, or other statistical measures.



NumPy: Numerical Computation

NumPy (Numerical Python) is a library that provides support for working with large, multi-dimensional arrays and matrices. It also includes a vast collection of mathematical functions to operate on these arrays.

Key Features:

N-dimensional Arrays: Efficiently handles large arrays with homogeneous data types.

Mathematical Functions: Perform operations like linear algebra, random number generation, and statistical analysis on arrays.

Vectorization: Allows you to perform operations on entire arrays without the need for explicit loops, speeding up computations.

Broadcasting: Enables operations on arrays of different shapes, making mathematical computations more flexible.

Matplotlib: Data Visualization

Matplotlib is a plotting library used to create static, animated, and interactive visualizations in Python. It allows you to generate a wide variety of plots such as

line graphs, bar charts, histograms, scatter plots, and more.

Key Features:

Customization: Provides extensive options for customizing the appearance of plots, including titles, labels, legends, and colors.

Multiple Plot Types: Supports a range of plot types to visualize different kinds of data.

Interactive Plots: Matplotlib can be used to create interactive plots that allow users to zoom in, pan, and explore data.

Figure Subplots: Allows the creation of multiple subplots within a single figure to compare multiple visualizations at once.

Seaborn: Statistical Data Visualization

Seaborn is a statistical data visualization library built on top of Matplotlib. It provides a high-level interface for creating aesthetically pleasing and informative visualizations.

Key Features:

Built-in Themes: Seaborn comes with several pre-designed themes and color palettes, which help create visually appealing plots with minimal effort.

Statistical Plots: Easily create complex plots like heatmaps, violin plots, pair plots, and box plots to explore relationships, distributions, and correlations in your data.

Integration with Pandas: Seaborn works seamlessly with Pandas Data Frames, allowing you to directly plot data stored in these structures.

Automatic Aggregation: Handles statistical aggregation of data, such as plotting means and standard deviations for grouped data.



Head function

The. head () function in Python is a method used with Pandas Data Frames and Series to quickly view the top few rows of a dataset. By default, it returns the first 5 rows of the Data Frame or Series

df.head(12)

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.0	220.0	Hair Oil & Serum	4.1	This Product contains Garlic Oil that is known...
1	2	Water Bottle - Orange	Kitchen, Garden & Pets	Storage & Accessories	Mastercook	180.0	180.0	Water & Fridge Bottles	2.3	Each product is microwave safe (without lid), ...
2	3	Brass Angle Deep - Plain, No.2	Cleaning & Household	Pooja Needs	Ttm	119.0	250.0	Lamp & Lamp Oil	3.4	A perfect gift for all occasions, be it your m...
3	4	Cereal Flip Lid Container/Storage Jar - Assort...	Cleaning & Household	Bins & Bathroom Ware	Nakoda	149.0	176.0	Laundry, Storage Baskets	3.7	Multipurpose container with an attractive desi...
4	5	Creme Soft Soap - For Hands & Body	Beauty & Hygiene	Bath & Hand Wash	Nivea	162.0	162.0	Bathing Bars & Soaps	4.4	Nivea Creme Soft Soap gives your skin the best...
5	6	Germ - Removal Multipurpose Wipes	Cleaning & Household	All Purpose Cleaners	Nature Protect	169.0	199.0	Disinfectant Spray & Cleaners	3.3	Slay protected from contamination with Multipu...
6	7	Multani Mati	Beauty & Hygiene	Skin Care	Satinance	58.0	58.0	Face Care	3.6	Satinance multani matti is an excellent skin t...
7	8	Hand Sanitizer - 70% Alcohol Base	Beauty & Hygiene	Bath & Hand Wash	Bionova	250.0	250.0	Hand Wash & Sanitizers	4.0	70%Alcohol based is gentle of hand leaves skin...
8	9	Biotin & Collagen Volumizing Hair Shampoo + Bi...	Beauty & Hygiene	Hair Care	SiBotanica	1098.0	1098.0	Shampoo & Conditioner	3.5	An exclusive blend with Vitamin B7 Biotin, Hyd...
9	10	Scrub Pad - Anti- Bacterial, Regular	Cleaning & Household	Mops, Brushes & Scrubs	Scotch brite	20.0	20.0	Utensil Scrub-Pad, Cloves	4.3	Scotch Brite Anti- Bacterial Scrub Pad th...

Data cleaning

is null ():

The `.is null ()` function is used to identify missing values in a Data Frame or Series. It returns a Data Frame or Series of the same shape.

sum ():

The `.sum ()` function, when applied to the result of `.is null ()`, counts the number of True values (i.e., the number of missing values).

A screenshot of a Jupyter Notebook interface. The top bar shows navigation icons. The code cell contains the command `df.isnull().sum()`. The output is a Series with the following values: index (0), product (1), category (0), sub_category (0), brand (1), sale_price (6), market_price (0), type (0), rating (8636), and description (115). The dtype is int64.

```
df.isnull().sum()
```

index	0
product	1
category	0
sub_category	0
brand	1
sale_price	6
market_price	0
type	0
rating	8636
description	115

dtype: int64

Interpretation

No missing values: Columns like index, category, subcategory, market price, and type have no missing values.

Few missing values: Columns like product, brand, sale price, rating, and description have some missing values. For instance, product and brand each have 1 missing value, while sale price has 6 missing values.

Many missing values: The rating column has a significant number of missing values (8636), and description has 115 missing values.

Once you've identified the missing values in your dataset, you can decide how to handle them. Some common strategies for dealing with missing data include:

Handling missing values

Imputation:

Replace missing values with a computed value, such as the mean, median, or mode of the column.

Removal:

Drop rows or columns with missing values, depending on the size of the dataset and the importance of the missing data.

Analysis:

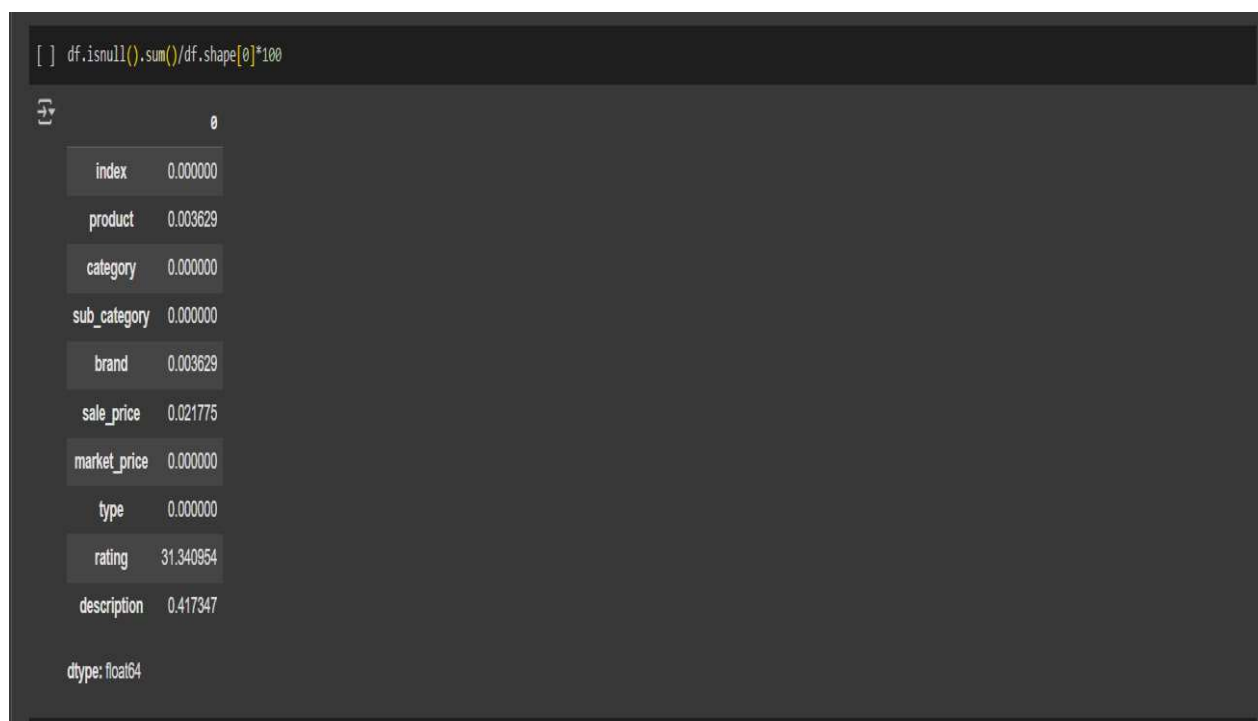
Sometimes, missing data may be informative (e.g., missing ratings could mean a product has not been rated). You may decide to keep these rows as they are or flag them for further investigation.

Forward/Backward Filling:

For time-series data or sequential data, you might fill missing values with previous or next valid entries.

percentage

```
[ ] df.isnull().sum()/df.shape[0]*100
```



	0
index	0.000000
product	0.003629
category	0.000000
sub_category	0.000000
brand	0.003629
sale_price	0.021775
market_price	0.000000
type	0.000000
rating	31.340954
description	0.417347

dtype: float64

The expression **`df.isnull().sum()/df.shape[0]*100`** is used to calculate the percentage of missing (null or Nan) values in each column of a Data Frame in Pandas. This percentage helps you quickly assess the extent of missing data in your dataset. It's particularly useful when:

- You want to determine whether a column has a significant amount of missing data that might affect your analysis.
- You want to decide on appropriate handling strategies (e.g., imputation, removal) based on the percentage of missing values.

Conclusion:

By calculating the percentage of missing values using

`df.isnull().sum()/df.shape[0]*100`, you gain valuable insight into the completeness of your data. This helps you make informed decisions about how to handle missing data, whether through imputation, removal, or other methods, ensuring that your analysis or machine learning models are based on clean, high-quality data.

-Filling missing values

Product:

```
[ ] df_1 = df[df['product'].isnull()]
df_1
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
14363	14364	NaN	Beverages	Coffee	Cothas Coffee	200.0	240.0	Ground Coffee	4.2	Cothas Specialty Blend Coffee and Chicory incl...

```
[ ] #filling missing values of product column by its original values.
df_1.loc[df_1.index[0], 'product'] = ('speciality blend')
```

Filling the missing value of product column through its original value because product has only 1 null value which easily we can find out

Reset index

```
[ ] df_1
```



	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
14363	14364	speciality blend	Beverages	Coffee	Colthas Coffee	200.0	240.0	Ground Coffee	4.2	Colthas Specialty Blend Coffee and Chicory incl...

```
df = df.set_index('index')
df_1 = df_1.set_index('index')

df.update(df_1)

df = df.reset_index()
```

- **Set the 'index' column as the index** of both df and df_1.
- **Update df with values from df_1** wherever the index labels match.
- **Reset the index** of df, returning it to the default integer-based index and converting the custom index back to a regular column.

Description:

Filling the null values of description by not mentioned

Rating:

Filling the null values of rating column by median

```
[ ] #filling the missing values of description by not_mentioned  
df['description'].fillna('not_mentioned',inplace=True)
```

```
[ ] #filling the missing value of rating column by median  
df['rating'].fillna(df['rating'].median, inplace=True)
```

df.isnull().sum()

	0
index	0
product	0
category	0
sub_category	0
brand	1
sale_price	6
market_price	0
type	0
rating	0
description	0

dtype: int64

```
[ ] df[df['brand'].isnull()]
df
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.00	220.0	Hair Oil & Serum	4.1	This Product contains Garlic Oil that is known...

Refresh data frame

Only sale price and brand null values left

```
df[df['brand'].isnull()]
df
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.00	220.0	Hair Oil & Serum	4.1	This Product contains Garlic Oil that is known...
1	2	Water Bottle - Orange	Kitchen, Garden & Pets	Storage & Accessories	Mastercook	180.00	180.0	Water & Fridge Bottles	2.3	Each product is microwave safe (without lid), ...
2	3	Brass Angle Deep - Plain, No.2	Cleaning & Household	Pooja Needs	Ttm	119.00	250.0	Lamp & Lamp Oil	3.4	A perfect gift for all occasions, be it your m...
3	4	Cereal Flip Lid Container/Storage Jar - Assort...	Cleaning & Household	Bins & Bathroom Ware	Nakoda	149.00	176.0	Laundry, Storage Baskets	3.7	Multipurpose container with an attractive desi...
4	5	Creme Soft Soap - For Hands & Body	Beauty & Hygiene	Bath & Hand Wash	Nivea	162.00	162.0	Bathing Bars & Soaps	4.4	Nivea Creme Soft Soap gives your skin the best...
...	...	...	...	...	...	...	...	...	...	...
27550	27551	Wottagirl! Perfume Spray - Heaven, Classic	Beauty & Hygiene	Fragrances & Deos	Layerr	199.20	249.0	Perfume	3.9	Layerr brings you Wottagirl! Classic fragrant b...
27551	27552	Rosemary	Gourmet & World Food	Cooking & Baking Needs	Puramate	67.50	75.0	Herbs, Seasonings & Rubs	4.0	Puramate rosemary is enough to transform a dis...
27552	27553	Peri-Peri Sweet Potato Chips	Gourmet & World Food	Snacks, Dry Fruits, Nuts	FabBox	200.00	200.0	Nachos & Chips	3.8	We have taken the richness of Sweet Potatoes (...)
27553	27554	Green Tea - Pure Original	Beverages	Tea	Tetley	396.00	495.0	Tea Bags	4.2	Tetley Green Tea with its refreshing pure, ori...
27554	27555	United Dreams Go Far Deodorant	Beauty & Hygiene	Men's Grooming	United Colors Of Benetton	214.53	390.0	Men's Deodorants	4.5	The new mens fragrance from the United Dreams ...

Brand:

Find out the null values in brand

Filling the missing value of brand column through its original value

```
[ ] #filling the missing value of brand column through its original value
df['brand'].fillna('clirik',inplace=True)
```

```
df.isnull().sum()
```

	0
index	0
product	0
category	0
sub_category	0
brand	0
sale_price	6
market_price	0
type	0
rating	0
description	0

dtype: int64

Sale price:

```
df.isnull().sum()

index 0
product 0
category 0
sub_category 0
brand 0
sale_price 6
market_price 0
type 0
rating 0
description 0

dtype: int64

[ ] df['sale_price'].isnull().sum()

6
```

So here is the total no of the null values of sale price column

Filling the null or missing values of sale price column by fill na method

```
[ ] df['sale_price'].isnull().sum()

6

df[df.sale_price.isnull()]

index product category sub_category brand sale_price market_price type rating description
1719 1720 Puja Flower Wicks - Puvvu Cleaning & Pooja Needs Jaji NaN 20.0 Camphor & Jaji Puja Flower Batti / Puvvu
Vathulu Batti Household Vathulu is made...
1720 1721 Powder - Sambar Foodgrains, Oil & Masalas & Spices Annapooma NaN 46.0 Blended Masalas Annapoma Spices are the
Masala preserve for Connoiss...
1721 1722 Steel Fork - Medium, Premium Kitchen, Garden & Crockery & Cutlery BB Home NaN 299.0 Cutlery, Spoon & BB Home provides fine and
Excel Series, BBCL08 Pets Fork Series.median of 0 4.1n1... classy cutlery that ...
1722 1723 Snack Mix - Dhokla Snacks & Branded Ready To Cook & Eat MTR NaN 65.0 Breakfast & MTR Dhokla is extremely
Foods special in Gujarat. It...
2401 2402 Battery AA 3UT Hi Top Kitchen, Garden & Appliances & Nippo NaN 150.0 Battery & This Nippo battery incorporates
Pets Electrical state-of-the-a...
2402 2403 Klassic Plain Cocktail Napkins Cleaning & Disposables, Origami NaN 32.0 Toilet Paper 3.9 Klassic Plain Cocktail Napkins
(22 X 22 cm) Household Garbage Bag (22 X 22 cm) 10...

[ ] df.sale_price = df.sale_price.fillna(df.market_price)

[ ] df.sale_price.isnull().sum()

0
```

Information of dataset

Now again we are checking the null values in the columns after handling it i.e. if any of the null values left in the column, we will fill those values

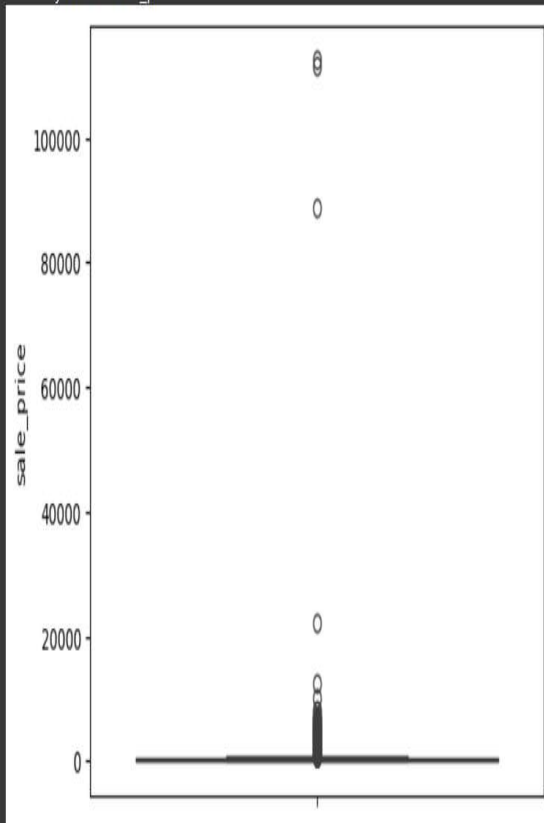
```
[ ] df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 27555 entries, 0 to 27554  
Data columns (total 10 columns):  
#   Column      Non-Null Count  Dtype  
---  ---  
0   index       27555 non-null  int64  
1   product     27555 non-null  object  
2   category    27555 non-null  object  
3   sub_category 27555 non-null  object  
4   brand       27555 non-null  object  
5   sale_price  27555 non-null  float64  
6   market_price 27555 non-null  float64  
7   type        27555 non-null  object  
8   rating      27555 non-null  object  
9   description  27555 non-null  object  
dtypes: float64(2), int64(1), object(7)  
memory usage: 2.1+ MB
```

Handling outliers

```
sns.boxplot(df.sale_price)
```

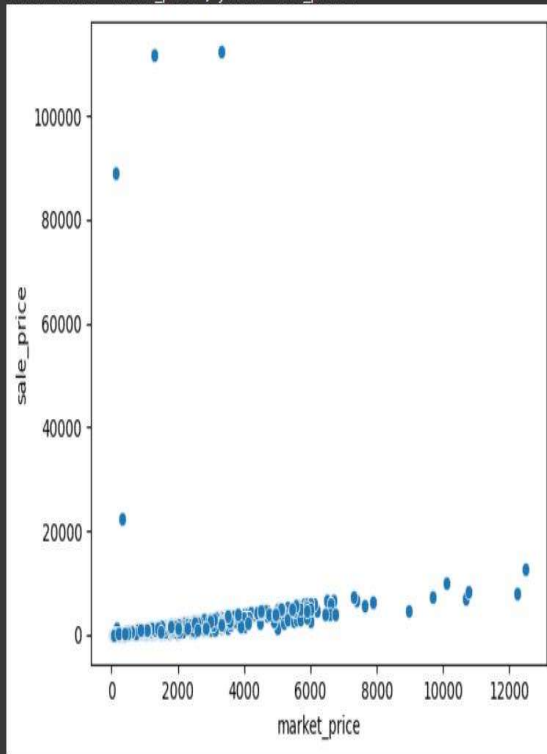
```
<Axes: ylabel='sale_price'>
```



```
[ ] df[df.sale_price>20000]
```

```
sns.scatterplot(y="sale_price",x="market_price",data=df)
```

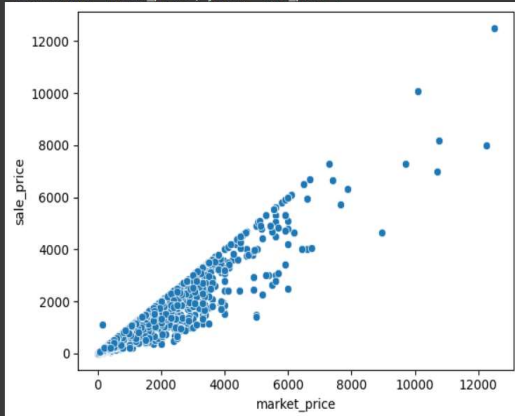
```
<Axes: xlabel='market_price', ylabel='sale_price'>
```



```
df.sale_price = np.where(df.sale_price>20000,df.market_price,df.sale_price)
```

```
[ ] sns.scatterplot(y="sale_price",x="market_price",data=df)
```

```
<Axes: xlabel='market price', ylabel='sale price'>
```

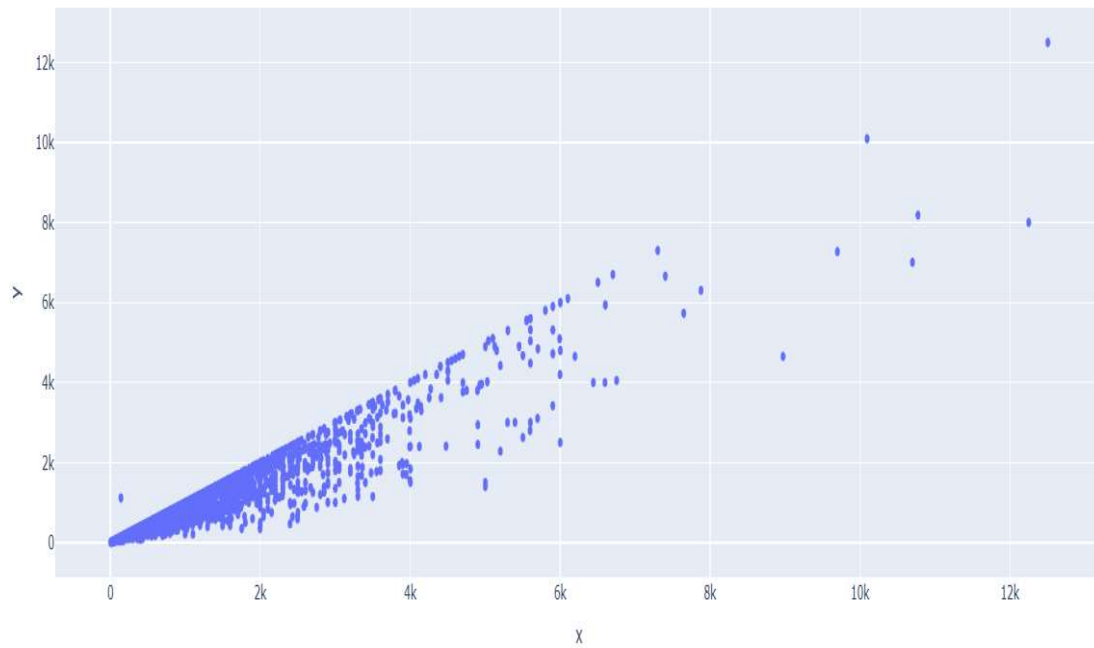


Sale price vs market price

```
#scatter graph through plotly  
px.scatter(y=df.sale_price,x=df.market_price,labels=True,title="sale_price vs market_price")
```

+

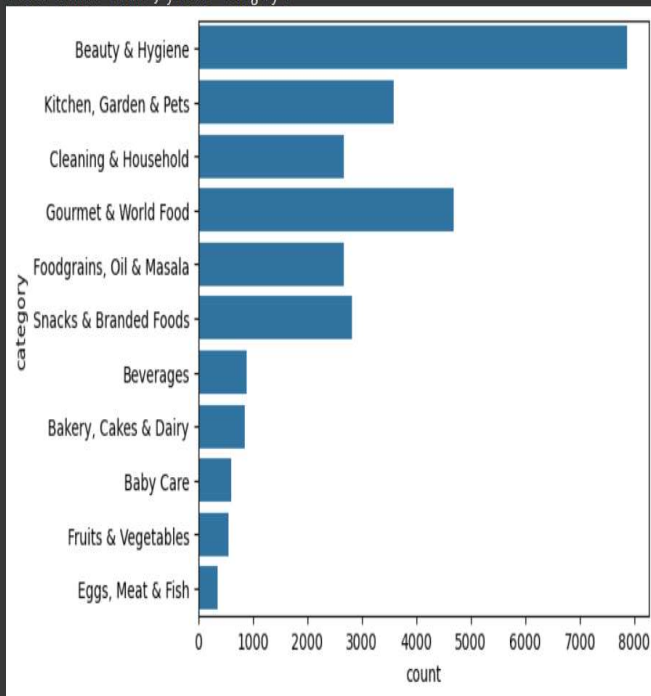
sale_price vs market_price



Visualizations

```
[ ] sns.countplot(y=df.category)
```

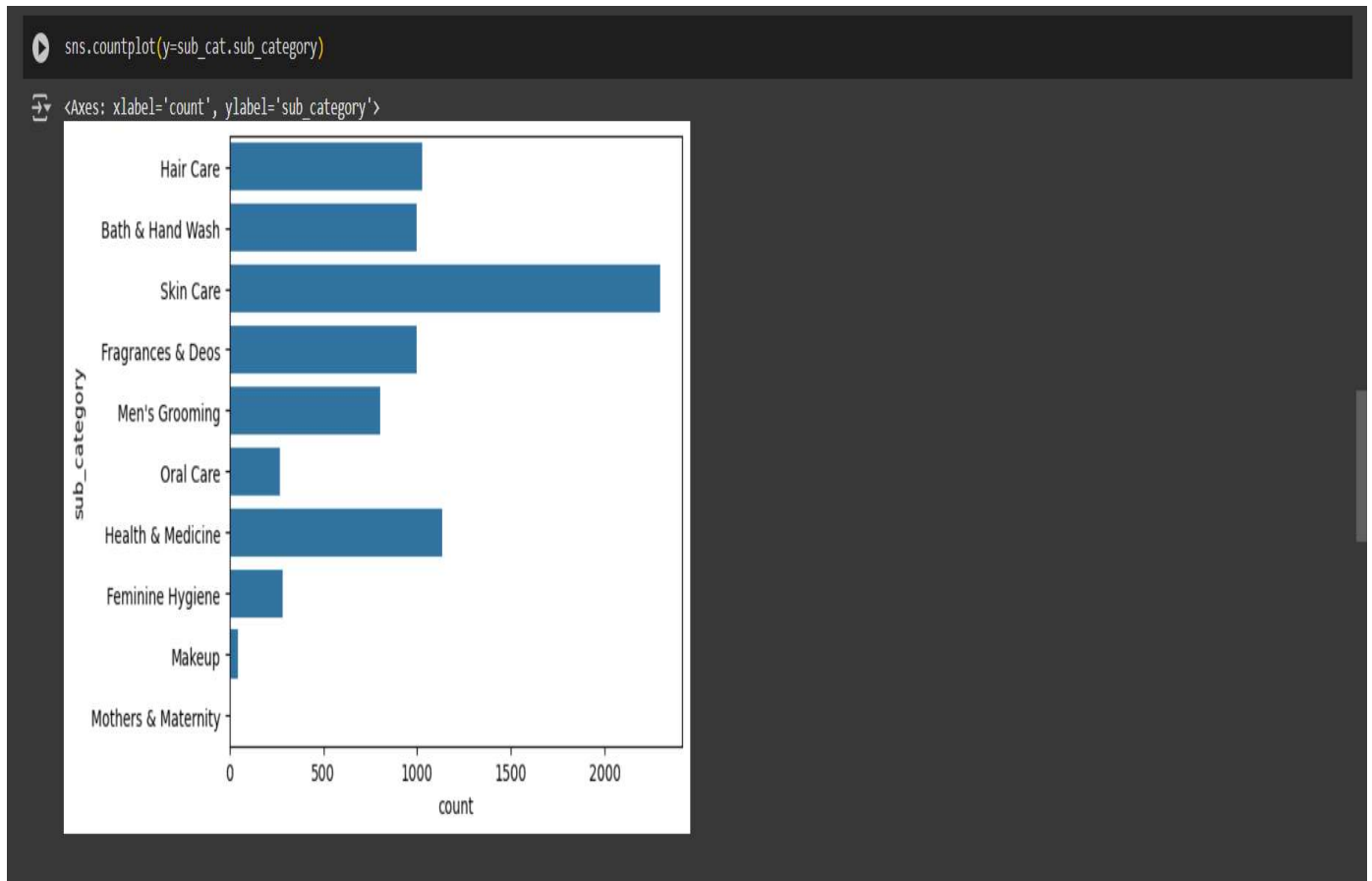
```
<Axes: xlabel='count', ylabel='category'>
```



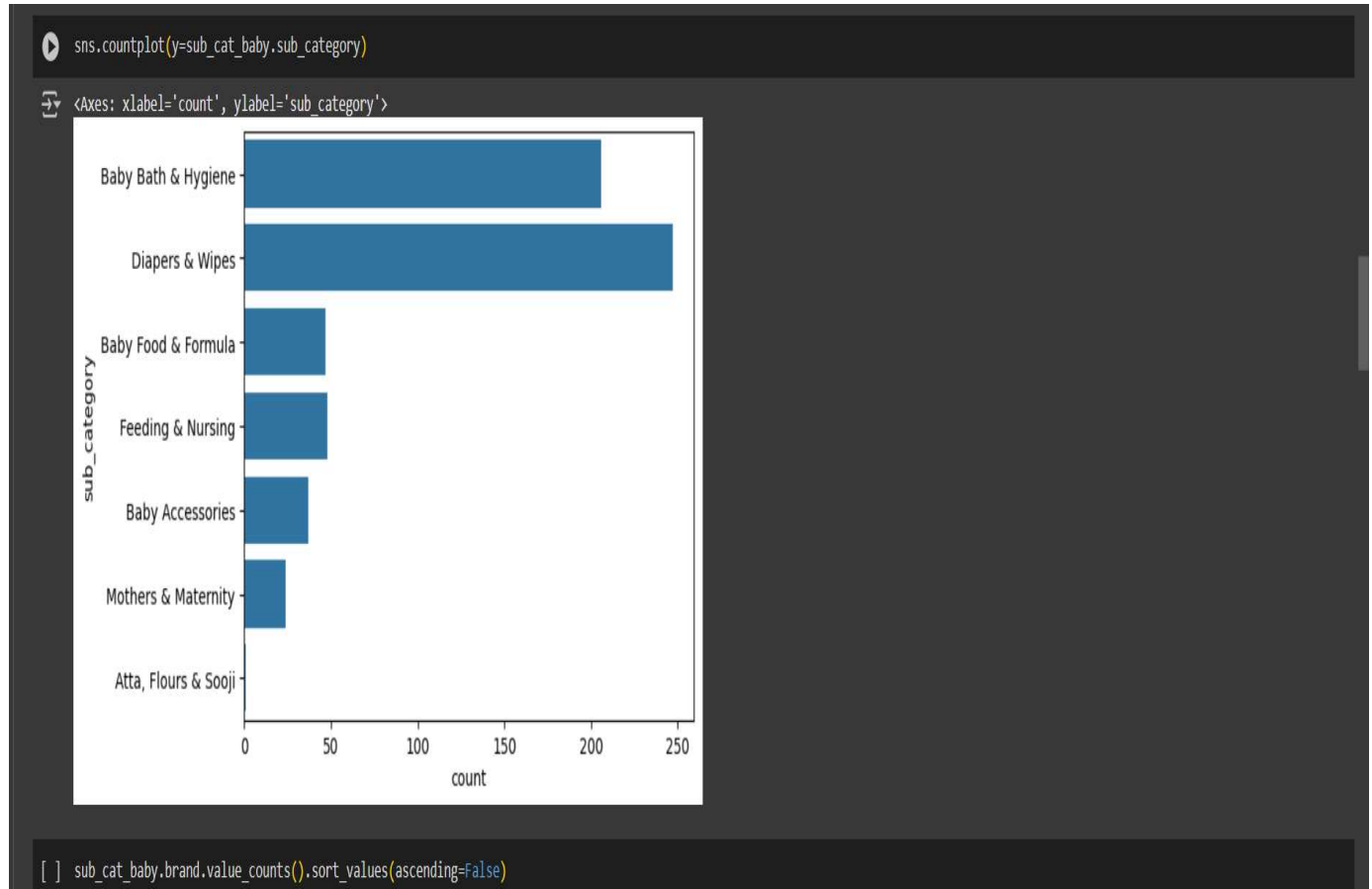
Conclusion

The product quantity analysis provides clear insights into consumer behaviour, category demands, and operational considerations. Categories with longer shelf lives or essential nature (e.g., Beauty & Hygiene, Kitchen & Pet Goods) tend to stock higher product counts. Conversely, categories with more specialized, perishable, or niche market demands, like Baby Care, Fruits & Vegetables, and Eggs, Meat & Fish, have fewer products. The surprising lower product range for Big Basket warrants a deeper investigation into their supply chain and strategy.

analysing the best-selling category specially beauty & hygiene



Least sold product i.e. baby care



The lower quantity of baby products could be attributed to their shorter period of usage and customer sensitivity

- . Diapers and wipes have higher quantities due to their daily use nature and increased market competition.
- . Baby bath and hygiene products are extensive quantity of products, indicating their frequent usage, wide range of products.

. The "Mothers & Maternity" category comprises 24 items, whereas in the "Beauty and Hygiene" category, there are 7 items. This discrepancy may suggest a greater specificity towards baby-related products in the former and a focus on maternal hygiene in the latter.

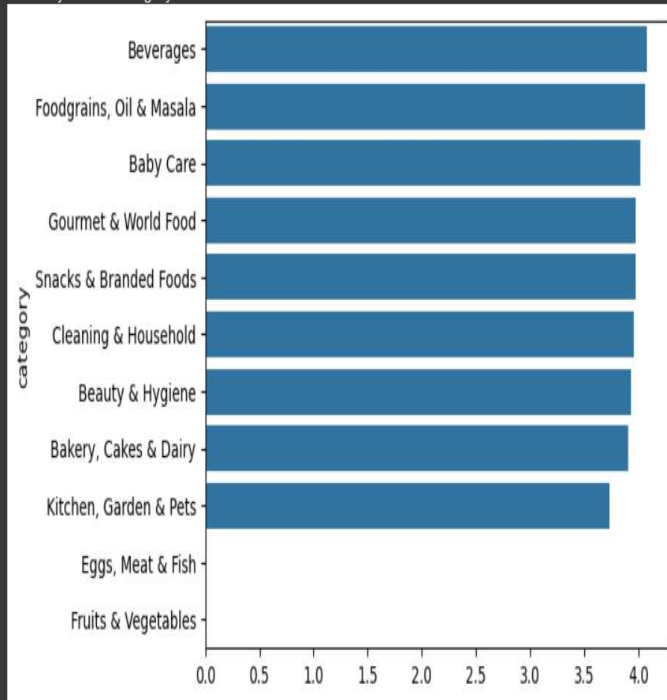
. The market is highly competitive, with Huggies and Pampers as dominant players. Anticipated future performance suggests that brands like Mee Mee, Himalaya, and The Moms Co. may thrive in a more competitive market, emphasizing quality products.

Analysis based on rating

-rating of the product

```
sns.barplot(x=data.values,y=data.index)
```

<Axes: ylabel='category'>



Decision taken based on the discounts offered

Reviewing items based on discount offered

Reviewing items based on the discounts offered.

```
df.groupby('category')['market_price'].sum()
```

category	market_price
Baby Care	364020.00
Bakery, Cakes & Dairy	134357.00
Beauty & Hygiene	3882642.22
Beverages	240927.00
Cleaning & Household	701150.35
Eggs, Meat & Fish	114042.42
Foodgrains, Oil & Masala	615833.00
Fruits & Vegetables	35889.55
Gourmet & World Food	1680993.95
Kitchen, Garden & Pets	2361574.40
Snacks & Branded Foods	396141.50

dtype: float64

```
[ ] # Assigning the Maximum Retail Price (MRP) to the variable "MRP" for ease of reference.
```

```
MRP = df.groupby('category')['market_price'].sum()
```

```
df.groupby('category')['sale_price'].sum()
```

↕

	sale_price
--	------------

category	
----------	--

Baby Care	326317.17
-----------	-----------

Bakery, Cakes & Dairy	121525.14
-----------------------	-----------

Beauty & Hygiene	329459.44
------------------	-----------

Beverages	212186.67
-----------	-----------

Cleaning & Household	605043.09
----------------------	-----------

Eggs, Meat & Fish	101114.12
-------------------	-----------

Foodgrains, Oil & Masala	516916.23
--------------------------	-----------

Fruits & Vegetables	28345.36
---------------------	----------

Gourmet & World Food	1500115.31
----------------------	------------

Kitchen, Garden & Pets	1817723.12
------------------------	------------

Snacks & Branded Foods	364675.08
------------------------	-----------

dtype: float64

```
[ ] # Assigning the Selling Retail Price (SRP) to the variable "SRP" for ease of reference.
```

```
SRP = df.groupby('category')['sale_price'].sum()
```

```
[ ] # Calculating discounts for each category using the Maximum Retail Price (MRP) and the Selling Price (SRP).
```

```
Discount = (MRP-SRP)/MRP * 100
```

```
[ ] Discount= Discount.round(2)
```

```
[ ] Discount = Discount.sort_values()
```

```
[ ] Discount
```

Calculating discount for each column

Discount



0

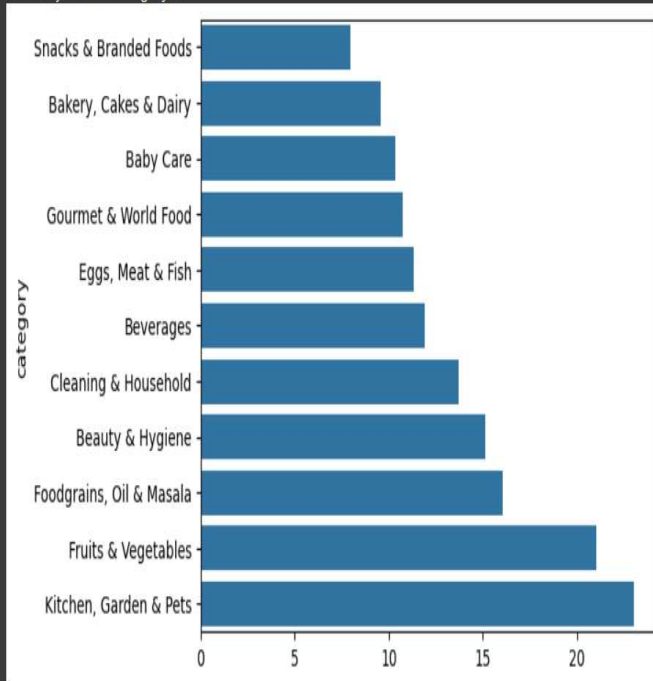
category

Snacks & Branded Foods	7.94
Bakery, Cakes & Dairy	9.55
Baby Care	10.36
Gourmet & World Food	10.76
Eggs, Meat & Fish	11.34
Beverages	11.93
Cleaning & Household	13.71
Beauty & Hygiene	15.15
Foodgrains, Oil & Masala	16.06
Fruits & Vegetables	21.02
Kitchen, Garden & Pets	23.03

dtype: float64

```
[ ] sns.barplot(y=Discount.index ,x=Discount.values)
```

↔ <Axes: ylabel='category'>



observation

- . The category "Kitchen, Garden & Pets" has the highest discount of 23.03%, likely due to high margin, clearance sales, or seasonal discounts to boost sales and attract customers in this segment.
- . The category "Snacks & Branded Foods" has the least discount at 7.94%, possibly because of thinner profit margins in this competitive market.
- . The higher discount rate of 21.02% in "Fruits & Vegetables" may be influenced by factors such as perishability, seasonal fluctuations, and the need to manage inventory with shorter shelf life

Thank you

Project link:

https://colab.research.google.com/drive/1fOzAqq9NLG7t8xZ4eyOSuLdakk_UB_z4#scrollTo=unh72kebeMAI

